

PARTS LIST

Semiconductors:

Board1	- Arduino Uno board
IC1	- 7812, 12V voltage regulator
BR1	- 1A bridge rectifier
D1	- 1N4007 rectifier diode
T1	- 2N2222 NPN transistor
LED1, LED2	- 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1	- 2.2-kilo-ohm
R2, R3	- 1-kilo-ohm

Capacitors:

C1	- 1 μ F, 25V electrolytic
C2	- 1000 μ F, 25V electrolytic

Miscellaneous:

RL1	- 12V, single-changeover relay
P1	- 12V DC fluid pump
SEN18	- 3-pin water-level sensor
	- HC-SR04 ultrasonic sensor
X1	- 230V AC primary to 15V, 2A secondary transformer
CON2	- 3-pin female Berg connector for SEN18
CON3	- 4-pin femal Berg connector for HC-SR04
	- 8-pin femal Berg connector for Arduino
	- 2-pin connector for X1 secondary transformer
	- General-purpose PCB
	- Jumper wire

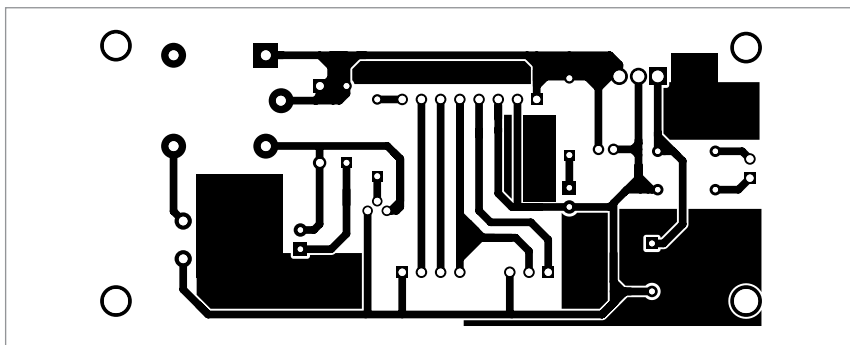


Fig. 2: Actual-size PCB layout for water dispenser

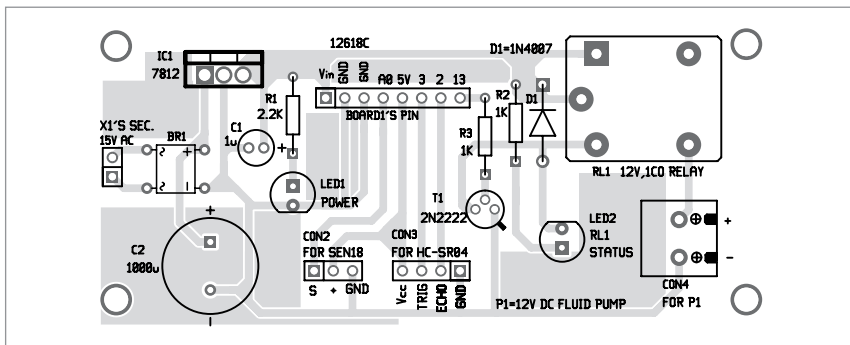


Fig. 3: Component layout of the PCB

Construction and testing

An actual-size PCB layout for the water dispenser circuit is shown in Fig. 2 and its component layout in Fig. 3. You can also use a general-purpose PCB to assemble the circuit.

After assembling the circuit, connect 15V secondary voltage from the transformer to the PCB across X1 marked in Fig. 3. Connect primary of transformer to 230V AC mains through CON1 (not shown in component layout.). Do not forget to upload the *dispenser.ino* code to Arduino before switching on the circuit.

As shown in Fig. 1, signal pin (S) of water-level sensor is connected to analogue pin A0 of the Arduino Uno board and GND and +Vcc terminals are connected to ground and 5V pins, respectively. This water-level sensor will detect the water level in pet's water bowl.

The 12V SPDT relay RL1 is used at the output side. High output at pin 13 of Arduino triggers the relay via switching transistor T1 to start DC fluid pump P1. D1 is used as a flywheeling diode to protect the transistor from back EMF (electromotive force). Indicator LED2 is used to show

relay's status. Pump P1 is connected as load across relay RL1 between normally open (NO) and common (C) terminals with series connection of 12V DC power source.

You need to set a threshold water level in your pet's bowl using sensor SEN18. Therefore, water level sensor should be first calibrated according to your need before the final setup. You can do it by following below-mentioned two steps after powering on the system and uploading *calibration.ino* program into Arduino.

Step 1. Mark the water level in the bowl where you want it to be considered full.

Step 2. Check the value on the serial monitor and note the full-level value.

Now put that value in *dispenser.ino* code. Compile the code and upload it to Arduino. This will help the

system to work accurately.

After the whole circuit setup, turn power supply on. The ultrasonic sensor will continuously check whether your pet is near the bowl to drink water or not. When the pet comes near the bowl, the program checks the water level from the sensor's feedback signal. If the water level sensor detects low level, Arduino program gives high signal to digital I/O pin 13 of Arduino, which energises relay RL1 through transistor T1. So, pump P1 connected to normally-open (NO) contact of the relay switches on and starts pumping water into bowl. When water level reaches the desired level, the program gives low signal to digital output pin 13 of the Arduino, which de-energises relay RL1 and thus disconnects power supply and stops pump P1.

This is how clean and fresh water can be ensured for pet(s) automatically without any manual intervention. **EFY**

EFY Note

The source code of this project is available for free download at source.efymag.com



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